

Meeting Summaries of June 5

Date: June 5, 2026

Time: PDT 7:00 to 7:45pm

Attendees: Prof. Jan Ciborowski, Dr. Javed, and Feng Gu

Topic: Progress review of thesis analysis, interpretation of NMDS results, thesis writing, and timeline adjustment.

Questions, Feedbacks and Suggestions

- Prof. Jan: The NMDS plots should include **site codes** so that individual points can be traced back to the original samples and checked biologically.
- Prof. Jan: The ordination result is not enough by itself. Feng should identify which **taxa contribute most strongly** to each NMDS axis, especially the taxa separating reference and degraded poles.
- Prof. Jan: The recent **chemical-data cleaning** is acceptable for continuing the analysis. He will review the comments on unit conversions and provide feedback before Monday.
- Dr. Javed: The thesis writing should become **simpler and more direct**. The methods and story should be written in a form suitable for thesis chapters rather than website notes.

Main conclusion from the meeting

The current work still needs more interpretation and writing before it is ready for defence. Prof. Jan and Dr. Javed advised Feng to **revise the thesis timeline**, since a July or August defence is not realistic under the current progress.

- Prof. Jan and Dr. Javed: Feng should continue converting gallery and website materials into chapter-style PDFs and send them out for review section by section.

Action Items

- Feng will add **site codes** such as DR25 to the NMDS figures.
- Feng will summarize the **taxa contributions** to each NMDS axis and report which taxa characterize reference and degraded conditions.
- Feng will update the **thesis timeline spreadsheet** and revise the expected dates for completion, submission, and defence.

- Feng will rewrite website and gallery content into **chapter-style PDFs** and send completed sections to Prof. Jan and Dr. Javed for review.
- Feng will continue the **piecewise quantile regression** work and, if possible, check bootstrap or residual-bootstrap confidence intervals for extreme quantiles.
- Dr. Javed will start reading Feng's chapters and coordinate with Prof. Jan for a joint review.
- Prof. Jan will review Feng's notes on **data units and processing** and send feedback before Monday.

Summary of Discussion

- Feng reported recent progress on **chemical data cleaning**, PCA analysis, clustering, and environmental classification. He explained how inconsistent chemical units were checked and converted before continuing with the later analyses.
- Feng presented a 'stepwise like but actually not' **LDA classifier** trained on concordant or reference sites using three filtered environmental variables. The classifier performed well on concordant sites but less well on conflicting sites, which led to discussion of training and testing on different site subsets.
- Feng then showed **NMDS ordination** based on Bray-Curtis dissimilarity of invertebrate relative abundances. Prof. Jan clarified that this ordination reflects only the taxa composition and does not directly include environmental variables, so the next interpretation must focus on which organisms drive the separation.
- The meeting ended with discussion of **thesis progress and writing**. Prof. Jan and Dr. Javed judged that the project was closer to 50–60% complete than to the level needed for a summer defence, so Feng was advised to revise the schedule and keep sending written chapters for review.